JenaClimateForecasting

EC417 Introduction to Machine Intelligence - Term Project P22

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1. Problem and measured finding

We forecast air temperature 24 hours ahead from 120 hourly weather observations. The primary endpoint is test MAE in degrees Celsius. RMSE provides a secondary measure of large errors. A persistence baseline asks whether a learned model provides value beyond repeating the latest temperature.

Calendar features reduced GRU 24-hour test MAE by 0.094 degrees C (4.3%). The lowest mean 24-hour MAE in this experiment belongs to GRU_calendar.

The experiment compares persistence, a regularized linear model on flattened windows, a GRU, and the same GRU with cyclical calendar inputs. Every method uses the same forecast origins and temperature targets. Seeds 17, 42 and 147 measure training variability for the neural models.

Scope

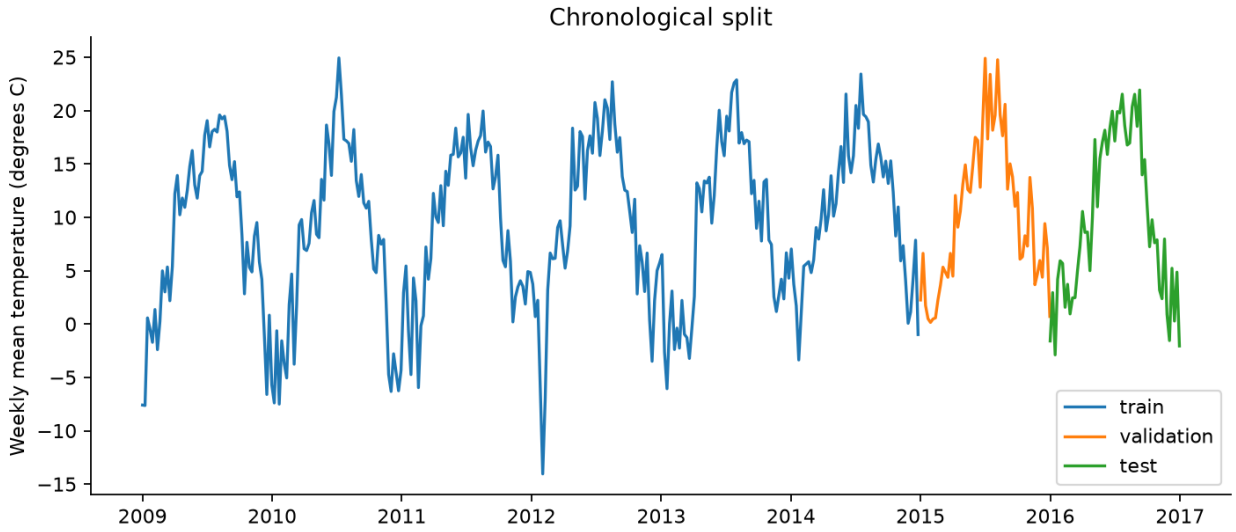
All results come from the supplied Jena 2009-2016 benchmark. The main endpoint is 24 hours; additional fixed horizons of 1, 6 and 12 hours show how forecast difficulty changes. All learned models share these outputs, making calendar inputs the isolated proposed improvement.

The design and implementation follow the P22 task and the general submission requirements. No target accuracy is assumed, and a proposed improvement can perform worse.

2. Data, EDA and leakage controls

The archive contains 420,551 records. The audit found 327 duplicate timestamps. Nonzero sentinel counts: wv (m/s): 18, max. wv (m/s): 20.

Raw observations split into 2009-2014 training, 2015 validation and 2016 test. Hourly means use bins ending at the timestamp. Wind speed/direction become Cartesian components before averaging. Missing inputs are forward-filled within a period and then filled with training medians. Feature means and standard deviations use training data alone.



Training has 17,467 windows at stride 3. Validation has 8,617 and test has 8,544 hourly windows. Each window contains 120 inputs. No window crosses a split boundary. Missing observed targets and missing persistence references are excluded.

The seasonal temperature cycle motivates cyclical features. The station download page states CC BY 4.0. The TensorFlow tutorial documents the exact historical mirror. Archive and CSV SHA-256 hashes, access timestamp and full source URLs are recorded in data/source_manifest.json.

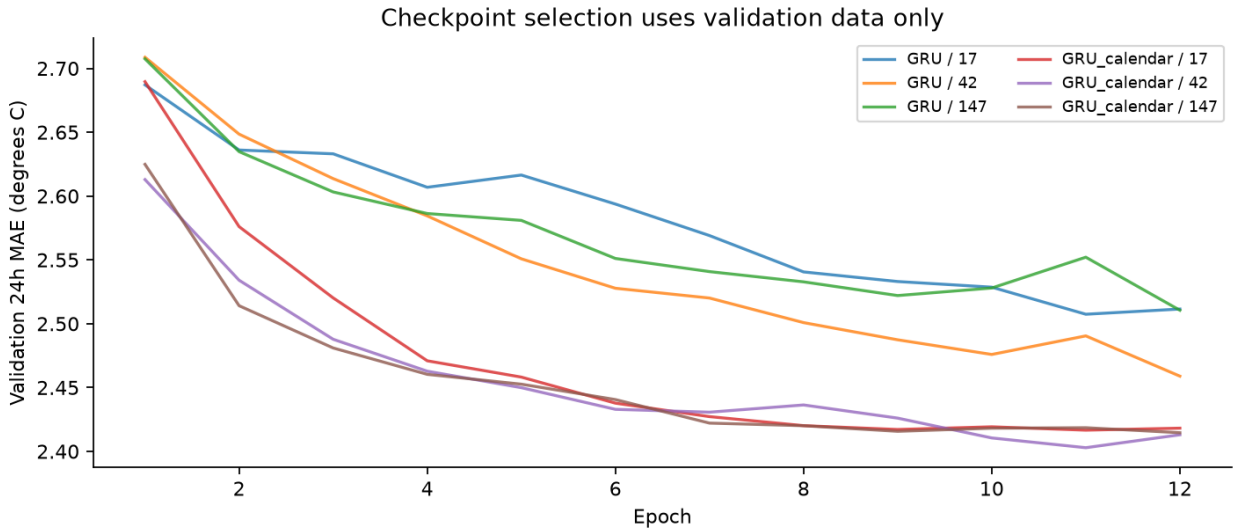
3. Models and controlled improvement

Persistence repeats the latest observed temperature. Ridge regression uses all 120 hourly inputs flattened into a vector; 2015 validation MAE selects alpha=1. The main model is a one-layer GRU with 24 hidden units and four linear outputs (1, 6, 12 and 24 hours). Both GRU variants use these same outputs, so the sole proposed improvement is adding sine/cosine of hour and day-of-year. Adam uses learning rate 0.003, batch size 256, up to 12 epochs, patience 3, and gradient clipping at 1. Training minimizes mean absolute error across the four standardized targets. Checkpoints minimize validation MAE at 24 hours. Normalization is reversed before reporting errors.

GRU: 2,980 trainable parameters. Selected epochs by seed: 17: 11, 42: 12, 147: 12.

GRU_calendar: 3,268 trainable parameters. Selected epochs by seed: 17: 11, 42: 11, 147: 12.

The calendar inputs are sin/cos of hour and sin/cos of within-year position, with leap years handled explicitly. They are bounded and deterministic. Main and improved models have equal hidden widths, output horizons, optimizer settings, data windows and seed lists. Their input parameter counts differ because four inputs are added.

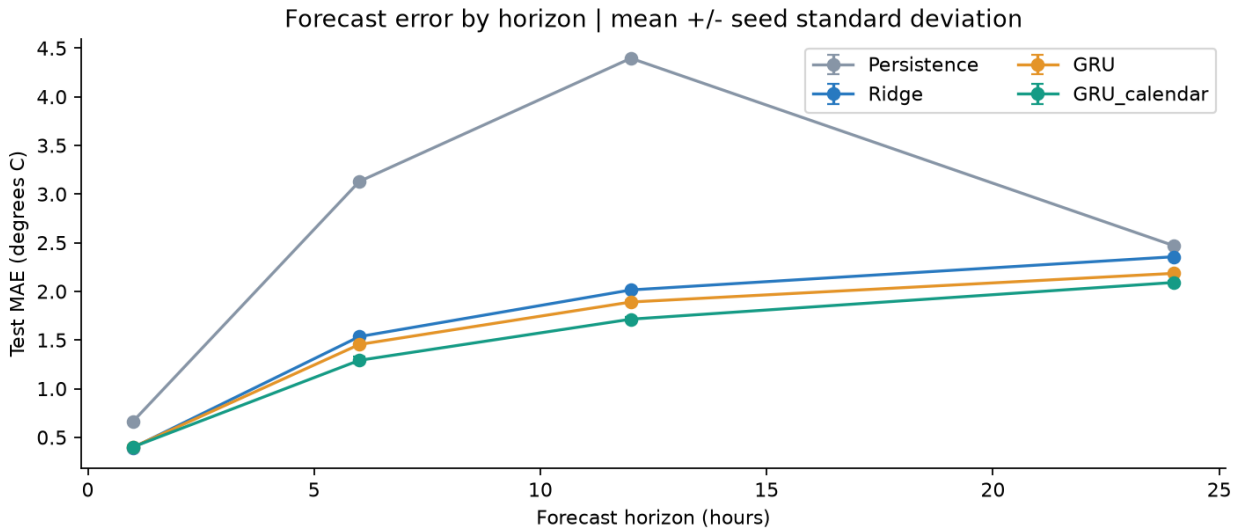


Learning curves show validation MAE at the primary 24-hour endpoint. Every run improves from its first epoch, and the selected checkpoints occur near the end of the fixed budget. This motivates testing longer training on validation data in future work, but does not prove it will help. Only validation results determine checkpoint choices.

4. Final test evaluation

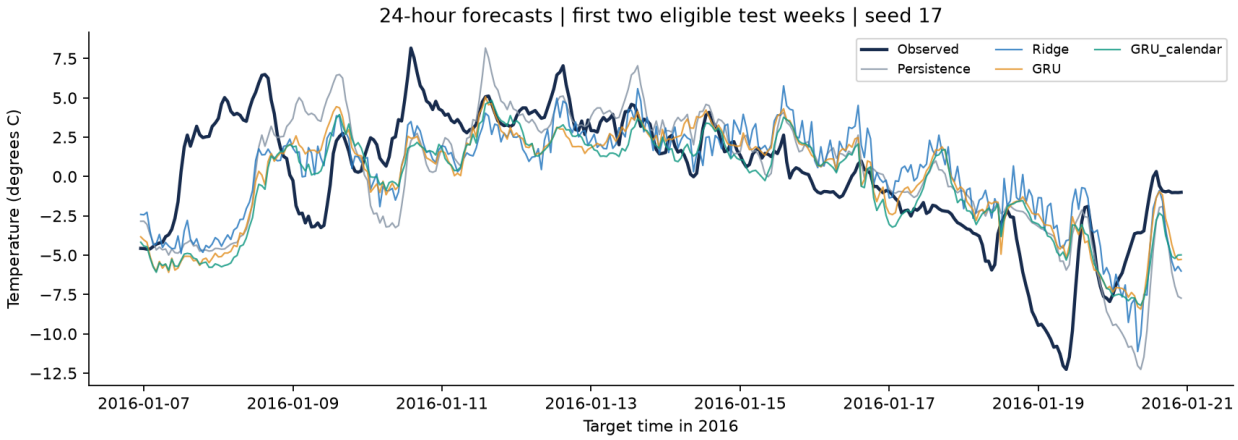
Model	MAE (C)	RMSE (C)	Time (s)
Persistence	2.469 +/- 0.000	3.237 +/- 0.000	0.0
Ridge	2.356 +/- 0.000	2.997 +/- 0.000	2.4
GRU	2.185 +/- 0.005	2.815 +/- 0.004	40.4
GRU_calendar	2.091 +/- 0.012	2.704 +/- 0.014	48.0

Neural results show mean +/- sample standard deviation across three seeds. Deterministic baselines run once and have zero seed variability. This does not establish zero uncertainty about future weather. Times include fitting and validation selection; neural times are per-seed means.



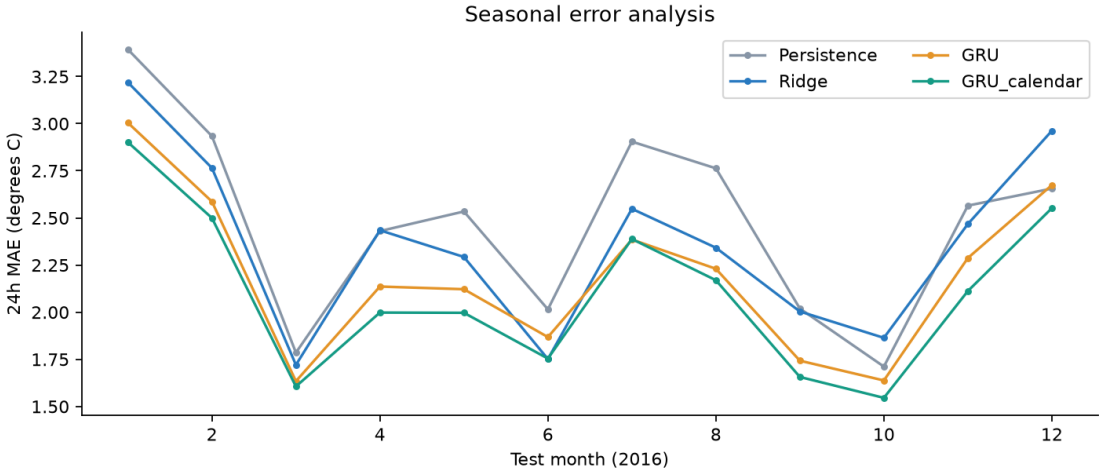
Calendar features reduced GRU 24-hour test MAE by 0.094 degrees C (4.3%). Persistence peaks at 12 hours and improves at 24 hours, consistent with the daily temperature cycle. This is a descriptive comparison, not a significance claim. Overlapping forecast origins violate independent-sample assumptions, so naive per-window confidence intervals would be misleading.

5. Forecast behaviour and error analysis



The chart uses the first eligible two test weeks and the predeclared seed 17. It compares individual forecasts, not a seed ensemble, at identical target times.

The calendar GRU's largest monthly mean error occurs in month 1 (2.901 C across 601 forecast origins). For seed 17, the largest stored error is 11.44 C at 2016-05-23 19:00:00: actual 13.19 C versus prediction 24.63 C. Rapid temperature changes and missing spatial weather information are possible explanations, not verified causes.



The monthly plot averages absolute errors over seeds. The largest-error table contains individual seed-17 examples. Weather-front explanations remain hypotheses without additional atmospheric or spatial observations.

6. Limitations, reproducibility and next steps

This experiment covers one station and one held-out year. Overlapping windows are correlated. Three-seed standard deviation quantifies optimization variability, not a confidence interval for future climates. Calendar features add a small number of input weights, so parameter counts are not exactly equal. Hourly averaging smooths extremes, imputation may distort unusual weather, and timestamps are used as provided without assuming a timezone conversion. The selected budget is modest and results do not establish the best possible GRU. This educational forecast is unsuitable for safety-critical weather alerts without further validation.

Next experiments should test variable-drop ablations and wider training budgets using validation data, then reserve a genuinely new final period. Rolling-origin evaluation would assess performance across multiple future periods. These are future work, not reported experiments.

Reproduction: Python 3.12; install requirements.txt, then run **python run_project.py**. Tests: **python -m unittest discover -s tests -v**. Measured platform: Windows-11-10.0.26300-SP0. PyTorch uses 2 CPU threads. Full configuration, validation selections, metrics and audit records are saved. Runtime depends on hardware and download speed.

AI-use statement

OpenAI Codex assisted with guideline interpretation, implementation, checks, experiments, plots, and draft documentation. Automated checks and execution are recorded in this project. The students must review the code and results, record their own verification and contributions, and be able to explain the work at the viva. This statement does not claim that student review has already occurred.

References

MPI weather data and terms. https://www.bgc-jena.mpg.de/wetter/weather_data.html

TensorFlow time-series tutorial and archive provenance.
https://www.tensorflow.org/tutorials/structured_data/time_series

Cho et al. (2014), Learning Phrase Representations using RNN Encoder-Decoder.
https://arxiv.org/abs/1406.1078

Kingma and Ba (2014), Adam. https://arxiv.org/abs/1412.6980

scikit-learn Ridge documentation.
https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.Ridge.html

Course sources: EC417_TermProjects.pdf (P22 and submission pages), EC417_Dataset_Guide.md, and updated allocation table. Source pages accessed 29 October 2026.